

VU KHANH CAO

Ottawa, ON ♦ +1 (343) 262-0178

vu.khanh.cao@gmail.com ♦ vucao.ca ♦ linkedin.com/in/vu-k-cao

EDUCATION

Carleton University, Ottawa, ON

Sep 2023 – Apr 2028

Bachelor of Computer Science (Honours) – Artificial Intelligence and Machine Learning

GPA: 3.75/4.00

EXPERIENCE

Software Developer Student

May 2026 – Present

Bank of Canada

Ottawa, ON

- Modernized Treasury Management Systems processing over **\$1 trillion in annual transactions** by developing Java Spring Boot services and TypeScript applications, and validating with Playwright.
- Built an agentic code-review suite integrated into Azure CI/CD pipelines to automatically enforce security and quality requirements across mission-critical systems, improving standards compliance by **60%**.
- Migrated legacy applications onto OpenShift-based cloud infrastructure, improving deployment reliability, scalability, and operational efficiency for systems supporting **2,000+ employees**.

Software Developer Student

May 2025 – Dec 2025

Ovenns Analytics Inc.

Ottawa, ON

- Built and shipped multi-tenant web and mobile platforms adopted by **10+ businesses**, owning development from product research and UX design through implementation, deployment, and production maintenance.
- Architected and developed **secure, scalable backend services** using **Node.js, Express, and PostgreSQL**, exposing **100+ API endpoints** while supporting multi-tenant access and reliable data workflows.

PROJECTS

FLIH – [Devpost](#)

Hack The North - Sep 2026

- Developed a **reinforcement learning pipeline** and simulation, reducing optimization step time from **8 minutes to 20 seconds** by minimizing CPU to GPU data transfers.
- Integrated the trained model into a **Jetson-based wheeled robot** with cameras, LiDAR, and building a web frontend to visualize training statistics and control the robot over bidirectional WebSockets.

JARVIS – [GitHub](#)

uOttaHack - Jan 2025

- Built a low-cost **physical-to-digital reconstruction pipeline** using QNX, Raspberry Pi, and Gaussian Splatting to produce web-ready 3D scenes from webcams, reducing capture hardware cost from **\$400 to \$20**.
- Containerized the reconstruction pipeline with **Docker** and integrated **AWS S3 storage with GPU accelerated processing** to automate image ingestion, model training, and web delivery of generated 3D scenes.

TRACY – [GitHub](#)

QHacks - Feb 2024

- Developed a **monocular computer vision pipeline** using TrackNetV2 and convolutional RNNs to reconstruct player and ball trajectories from a **single camera**, eliminating the need for specialized hardware.
- Built gameplay analytics for **serve speed, ball bounces, player heatmaps, and in/out detection** using Matplotlib and FFmpeg, contributing to a **3rd-place finish at QHacks 2024**.

SKILLS

Languages

Python, Java, JavaScript, TypeScript, C/C++, SQL

Libraries & Frameworks

React, Angular, Node.js, Express, Spring Boot, TensorFlow, OpenCV

Platforms & Technologies

Azure, AWS, Docker, Kubernetes, ArgoCD, OpenShift, REST APIs, Git